PORTFOLIO PROJECT

Beverages Ltd

Revenue Risk & Payment Intelligence Dashboard

BI Analyst | Power BI | Excel | SQL | Data Analysis

Total Revenue	Revenue at Risk	% Revenue at Risk	Avg Days to Pay	Total Units Sold
£1.21M	£40.79K	3.36%	31 days	2M

01 EXECUTIVE SUMMARY

This project delivers a six-page, fully interactive Power BI report built for Beverages Ltd, a multi-region beverage distributor operating across the West, Northeast, and South of the United States. The report centralises revenue performance, payment behaviour, product contribution, and retailer-level credit risk into a single, decision-ready intelligence platform.

The project spans the full analytics lifecycle from raw data extraction and validation through to insight delivery. SQL was used to extract and query transactional sales and payment data from the source database, ensuring a clean, structured dataset from the outset. Excel was then applied to validate, cross-check, and pre-process the data, confirming accuracy and consistency before ingestion into Power BI for modelling, DAX engineering, and dashboard development.

With £40.79K of revenue at risk representing 3.36% of total revenue and four major retail partners exhibiting payment cycles ranging from 30 to 65 days, the business required a structured, real-time view of its financial exposure. This dashboard provides exactly that: transforming raw transactional data into actionable insight for finance, operations, and commercial leadership.

02 BUSINESS CHALLENGE: THE LIQUIDITY GAP

Beverages Ltd faced a multi-layered cash flow risk problem without the tools to see it clearly. The core challenges were:

- No unified view of payment velocity across retailers, finance teams were working from fragmented spreadsheets with no benchmark comparison.
- Rising collection cycles average days-to-pay peaked at 34 days in January, with individual retailers breaching 65 days, creating significant liquidity pressure.
- Inability to isolate brand-level credit risk the business could not determine which beverage products were disproportionately driving payment delays.
- Geographic blind spots regional revenue concentration and risk exposure were unquantified, making sales strategy decisions largely intuitive rather than evidence-based.
- No early warning system without a structured aging profile or retailer accountability layer, at-risk receivables were only identified reactively, never proactively.

The business objective was clear: build a real-time, interactive BI solution that transforms payment and sales data into a predictive risk management tool enabling leadership to act before receivables age into bad debt.

03 TOOLS & TECHNICAL STACK

This project was executed across a three-tool analytics pipeline; SQL for extraction, Excel for validation, and Power BI for modelling and visualisation reflecting a professional, end-to-end BI workflow:

	Tool / Layer	Application in This Project
01	Excel *Data Extraction*	Used Excel as the primary tool for data extraction and initial data preparation, given the dataset format. I conducted data quality checks by identifying and handling missing values, and applied **VLOOKUP and XLOOKUP** to cross-reference retailer and product records against master reference tables, resolving inconsistencies at the source. This ensured the dataset was structured, clean, and ready for further validation.
02	SQL *Data Validation*	The cleaned dataset was exported into SQL for advanced validation and integrity assurance. I identified and removed **duplicate records**, verified data relationships, and consolidated datasets using **UNION operations** where necessary. Additionally, I applied **JOINs, WHERE clauses, and aggregate functions (SUM, AVG, COUNT, GROUP BY)** to validate key business metrics such as revenue totals and payment cycles ensuring the dataset was accurate, consistent, and fully analysis-ready before loading into Power BI.

03	**Power BI** *Modelling & Visualisation*	Built a star schema data model with a central fact table (sales & payments) linked to retailer, product, region, and date dimensions. Authored DAX measures for dynamic KPIs including Revenue at Risk, Payment Velocity benchmarking, risk tier classification, and aging profile percentages. Designed six interactive report pages with custom tooltips, drill-through navigation, Azure Maps integration, conditional formatting, and dynamic slicer filters.
04	**Power Query** *DAX | Azure Maps*	Power Query (M Language) used for final data transformation, type casting, and load optimisation into the Power BI model. DAX applied for all calculated measures and risk scoring logic. Microsoft Azure Maps embedded for geographic sales visualisation with interactive state-level drill-through and brand risk profile tooltips.

SQL Extract & Query → **Excel** Validate & Clean → **Power BI** Model & Visualise → **Insight** Deliver & Act

04 DASHBOARD ARCHITECTURE

The report is structured across six purposefully designed pages, each serving a distinct analytical function:

FRONT PAGE

- Navigation hub with branded design professional entry point connecting all six report sections.
- Sets the tone for stakeholder-facing delivery and mirrors a board-ready presentation format.

OVERVIEW

- **Headline KPIs:** Revenue (£1.21M), Revenue at Risk (£40.79K), Average Days to Pay (31, vs Benchmark -0.41), and % Revenue at Risk (3.36%).
- **Monthly Combo Chart:** Total Sales + Avg Days to Pay sales climb from £82K (Jan) to a peak of £116K (Aug), while payment days drop sharply from 34 → 30 by April and stabilise for the full remainder of the year.
- **Status Annotation:** An embedded green callout confirms the business outcome "Collections stabilized at 30 days post-Q1 policy enforcement."
- **Interactive Tooltip (Month-Level):** Hover over any month bar to reveal revenue, revenue at risk, retailer count, and peak payment delay. January surfaces £18.33K at risk across 4 retailers with a peak delay of 65 days the outlier that triggered the Q1 policy.

PRODUCTS

- **Brand Sales Volume Chart:** Coca-Cola leads with 461.9K units, followed by Diet Coke (413.75K), Dasani Water (392.75K), Sprite (384.7K), Fanta (338K), and Powerade (318.75K).
- **Revenue Contribution Treemap:** Dasani Water tops revenue share at 19.27%, with Coca-Cola (18.83%), Diet Coke (17.67%), Sprite (15.55%), Powerade (14.94%), and Fanta (13.73%).
- **Product Payment Velocity Table:** Three risk tiers revealed — Diet Coke at 65 days (HIGH), Sprite at 57 days (MEDIUM-HIGH), and standard 30-day compliant brands flagged LOW.
- **Brand-Level Tooltip (Risk Driver):** Hovering on any bar fires a custom tooltip identifying the Primary Risk Driver retailer Coca-Cola surfaces Costco with £3.88K fully at risk and a Collection Velocity gauge reading 65 days.

RETAILERS

- **Credit Risk Scatter Plot:** Maps all retailers by sales volume vs. average payment days. A 31-day risk threshold line separates stable performers from credit concerns Walgreens and Monitor hover above benchmark.
- **Revenue at Risk Table:** Target carries the highest exposure at 17.08% of its revenue at risk, followed by Walgreens (16.27%), Costco (12.29%), and Walmart (10.41%).
- **Dynamic Slicers:** Very Late / On Time / Late Payment filters instantly reconfigure the scatter plot and risk table, enabling rapid retailer segmentation.

GEOGRAPHY

- **Regional Revenue Bar Chart:** West region dominates at £582.40K, followed by Northeast (£391.77K) and South (£238.85K) with a risk gradient colour scale overlaid at 3.36% average.
- **Azure Map (Interactive):** Bubble markers in California, Texas, and New York allow click-level drill-down. New York reveals a 360-day collection anomaly and a Brand Risk Profile breakdown showing Late vs On Time exposure by product.
- **Payment Filter Buttons:** Very Late / On Time / Late payment toggles re-filter both the bar chart and map simultaneously.

RETAILER DETAILS

- **Payment Velocity vs. Benchmark Chart:** Collection days trend from 34 (Jan) down to 30 (Apr) and hold flat through December with the Q1 policy enforcement annotation confirmed on-chart.
- **Liquidity Matrix Scatter:** Brands plotted by £ volume and payment behaviour. Sprite and Diet Coke fall in the Critical Risk zone; Coca-Cola and Dasani Water sit near the benchmark threshold.
- **Company-Wide Aging Profile:** 96.64% of receivables are on time a strong top-line figure that the Accountability table contextualises with precision.
- **The Accountability Table:** Costco and Target flagged Very Late / High Risk / Score 3, with £7,750 and £2,362.50 at risk respectively. Total portfolio exposure: £40,787.50.
- **Layered Tooltips (Two-Tier):** January hover shows Aging Profile 13% Late / 78% On Time / 9% Very Late CAUTION: 3.0 days over target. Costco hover shows Actual Cycle 65 vs Target 31, Aging 100% red CRITICAL: 34 days behind schedule.

05 KEY INSIGHTS & FINDINGS

1

Q1 Policy Enforcement Delivered a Measurable Turnaround

Average days-to-pay dropped from 34 in January to 30 by April and held flat for 8 consecutive months. This is a documented, data-confirmed outcome not an estimate. The status annotation on the Overview chart provides an audit-ready record of the intervention's impact.

2

Costco is the Single Largest Risk Driver in the Portfolio

With a payment cycle of 65 days and 34 days beyond the 31-day target — and a 100% red aging profile, Costco represents a critical credit risk. Its exposure of £7,750 is the largest individual retailer balance in the accountability table.

3

Target Carries the Highest Proportional Revenue at Risk

Despite Costco's larger absolute exposure, Target's revenue-at-risk percentage of 17.08% is the highest in the portfolio, indicating a systemic payment behaviour problem that is disproportionate to its revenue contribution.

4

Diet Coke's 65-Day Payment Cycle Creates Product-Level Risk Concentration

At the product level, Diet Coke's average days-to-pay of 65 combined with Sprite at 57 days creates a dual-brand risk cluster that the standard retailer view would not reveal. Brand-level payment monitoring is not optional; it is essential.

5

The West Region Generates 47.9% of Total Revenue With Concentrated Risk

The West at £582.40K is the dominant revenue region, but New York's 360-day collection anomaly is an outlier that requires immediate forensic investigation. If left unaddressed, it could distort the regional performance picture and compound into a material receivables issue.

06 BUSINESS RECOMMENDATIONS

Escalate Costco to Credit Review Immediately [CRITICAL]

With a 65-day actual cycle against a 31-day target and 100% of its aging profile in the red, Costco must be escalated to a formal credit review. Payment terms should be renegotiated, and supply should be conditionally linked to settlement of the outstanding £7,750 balance.

Apply Payment Milestone Monitoring to Target and Walgreens [HIGH PRIORITY]

Target's 17.08% revenue-at-risk rate and Walgreens' 16.27% rate signal structural payment behaviour issues. Monthly milestone check-ins with account managers and automatic late payment alerts should be introduced within 30 days.

Monitor Coca-Cola's Late Payment Flag in New York [MEDIUM PRIORITY]

While New York maintains a healthy 31-day collection cycle overall, Coca-Cola is the sole brand flagging a Late payment status in the region's Brand Risk Profile. A targeted review of the Coca-Cola–New York retailer relationship should be conducted to determine whether this is an isolated incident or an emerging pattern before it escalates to a higher risk tier.

Embed Brand-Level Payment Velocity into Monthly Sales Reviews [ONGOING]

Diet Coke and Sprite's elevated payment cycles are not visible in standard retailer reporting. Including product-level payment velocity in monthly commercial reviews ensures credit risk is assessed at the most granular actionable level.

07 RESULTS & IMPACT

Metric	Outcome
Data Extraction via SQL	Structured queries joined retailer, product, and payment tables producing a single, clean analytical dataset with zero manual copy-paste and full traceability to source
Data Accuracy via Excel	Applied VLOOKUP/XLOOKUP for data reconciliation and cross-referencing, ensuring 100% row-level accuracy across all revenue and payment fields prior to Power BI ingestion.

Collection Cycle Reduction	34 days → 30 days (Jan–Apr); maintained for 8 months post-policy
Revenue at Risk Identified	£40,787.50 surfaced and attributed to specific retailers and products
Risk Visibility	100% of retailer exposure quantified across 3 payment tiers and 5 risk scores
Reporting Efficiency	Replaced fragmented spreadsheet reporting with a single 6-page self-service dashboard
Decision Speed	Retailer risk surfaced in under 2 clicks via custom tooltips and drill-through navigation
Geographic Intelligence	£1.21M revenue mapped and risk-scored across 3 regions and 3 states for the first time

08 STAKEHOLDER VALUE

This dashboard was engineered to serve multiple business functions simultaneously each stakeholder group deriving direct, role-specific value:

Stakeholder	Value Delivered
CFO / Finance Director	Real-time view of total revenue at risk (£40.79K), aging profile, and payment cycle benchmarking — enabling proactive treasury and working capital decisions without waiting for month-end reports.
Credit & Collections Team	Retailer-level accountability table with payment status, risk score, and amount at risk per partner. Custom tooltips surface critical alerts (e.g., Costco: 34 days behind schedule) enabling immediate, targeted collections action.
Commercial / Sales Leaders	Brand sales volume rankings, revenue contribution treemap, and retailer credit risk scatter plot providing a complete commercial performance picture that informs pricing, terms, and retailer prioritisation decisions.
Regional Sales Managers	Interactive Azure map with state-level drill-down and brand risk profiles by region enabling territory-specific performance reviews. New York's 32.30% revenue share and compliant 31-day cycle confirm it as a high-value, low-risk market, with Coca-Cola flagged as the sole brand requiring monitoring.
Product / Brand Managers	Product-level payment velocity data and revenue share analysis revealing that Diet Coke and Sprite carry disproportionate payment risk relative to their sales volume, informing product-level credit terms.

Executive Leadership	A single, board-ready overview page with KPI summary, monthly trend narrative, and policy impact confirmation designed for rapid consumption and strategic decision-making without requiring any technical BI knowledge.

This project demonstrates end-to-end BI delivery — from data modelling and DAX engineering to UX design, stakeholder storytelling, and business impact quantification.

The result is not just a dashboard — it is a decision intelligence platform.